

Data-driven fault-tolerant formation control for nonlinear quadrotors under multiple simultaneous actuator faults

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ABSTRACT

This paper addresses the problem of data-driven fault-tolerant formation control for quadrotors with nonlinearities, unknown system parameters, and multiple actuator faults in the vehicle dynamics. A distributed fault-tolerant formation control law is developed including a distributed observer to generate the position reference for each vehicle, a fault-tolerant position control law to track the position reference, and a fault-tolerant attitude control law to regulate the attitude. Reinforcement learning approaches are implemented to update the optimal control weights in the fault-tolerant formation control law design. Stability of the proposed fault-tolerant formation control law is proven and simulation results of quadrotors under multiple actuator faults are provided to demonstrate the effectiveness of the proposed method.

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1. Introduction

The formation flight of multiple unmanned aerial vehicles (UAVs) has attracted significant attentions in many practical application fields, such as cooperative communication relay, cooperative surveillance, and cooperative wilderness search (see, e.g., [1]). As a typical class of UAVs, the quadrotor systems are widely used for the capabilities of hovering and vertical take-off and landing. The cooperative interaction of multiple quadrotors provides greater efficiency and redundancy in a flight mission compared to a single quadrotor. However, the quadrotors are subject to actuator faults due to the degradation or damage of rotors. Furthermore, the actuators of multi-quadrotor systems are more prone to fault effects because of the enlarged team scale [2]. The potential actuator faults in the multi-quadrotor vehicle system will result in the degradation of the formation flight performance and even a chain of catastrophes [3]. Therefore, it is important to obtain a fault-tolerant control (FTC) method to achieve the formation of quadrotors with multiple actuator faults.

In general, the FTC methods can be categorized into the passive FTC methods and the active FTC methods. The passive FTC methods are popular for their low computation load, simplicity in algorithm structure, and robustness in small actuator faults (see, e.g., [4]). In the passive FTC approaches design, a fixed control

policy is given to restrain the predefined actuator faults based on robust control approaches such as robust compensation [5], H_∞ control [6], and sliding mode control [7]. However, the performance of the passive fault-tolerant controller usually degrades as the actuator faults increase [8]. On the other hand, the active FTC methods can guarantee a desirable performance when the actuator faults increase. In the active fault-tolerant control law design, the actuator faults are diagnosed and then controller parameters are reconfigured online. In [9], a nonlinear observer was applied to estimate the faults but the quadrotor dynamics was linearized to facilitate the controller design. In [10], the complete nonlinear quadrotor dynamics was considered and a sliding mode control (SMC) method was used for the attitude control of a quadrotor. However the formation problem of multiple quadrotors under multiple simultaneous actuator faults was not further studied in [10].

Recently, several FTC methods have been studied in the coordination of multi-agent systems, as shown in [11–14]. In [11], a dynamic surface control method was used in the FTC scheme to complete the distributed synchronization task of multi-UAV systems under actuator faults. In [12], the health conditions of the UAV team were considered and healthy/redundant actuators were applied to compensate the effects of actuator faults in the formation of multiple UAVs. In [13], an adaptive fault estimator was applied to estimate the actuator faults and a cooperative FTC law was designed for a multi-quadrotor system in the formation flight. In [14], a decentralized adaptive FTC method, integrated with disturbance observers, sliding-mode differentiators, and neural networks, was studied for the attitude synchronization

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flight of a multi-UAV system subject to actuator faults. It should be noted that the UAV models were required in the fault-tolerant cooperative control law design of [11–14] and the optimality of the designed controllers was not discussed. However, the multi-quadrotor system is a highly nonlinear vehicle system with multiple actuator faults, which makes it difficult for the traditional methods mentioned above to achieve the optimal cooperative control for heterogeneous multi-quadrotor system. Furthermore, as illustrated in [15], the payloads in each flight mission are usually unknown for the quadrotor team, which will lead to unknown quadrotor system parameters in practical applications. The unknown quadrotor system parameters make it challenging to guarantee the optimality of traditional model-dependent control methods for quadrotors in practical applications. Therefore, it is necessary to develop data-driven approach for multi-quadrotor system without requirement of quadrotor system parameters.

The reinforcement learning (RL) is an effective data-driven approach to learn the optimal control policies for vehicle systems with unknown dynamics (see, [16]). In [17], the RL algorithms were applied to the position and attitude subsystem of a quadrotor to restrain the time-varying and coupled uncertainties. In [18], the RL algorithm was used in the least-restrictive supervisory control protocol design for a quadrotor to achieve vertical trajectory tracking. In [19], a RL-based adaptive scheme was designed to plan construction tasks for a single quadrotor. In [20], multiple quadrotors were simplified as heterogeneous linear systems and RL-based optimal policies were used to achieve the attitude synchronization task. In [21], a robust formation control law including a robust position control law and a robust attitude control law was constructed for multiple quadrotors and the RL algorithms were designed to learn the optimal position and attitude policies. However, the literatures in [17–21] mainly focused on the optimal control of quadrotor systems and the effects of multiple actuator faults on the quadrotor control systems were not discussed. In [22], the RL algorithms were used to estimate the actuator faults and a FTC law was constructed for a single quadrotor with three degree-of-freedom (DOFs). In [23], the optimal attitude control policies were learned via RL algorithms and fault-tolerant synchronization controller was constructed for multiple quadrotors with three DOFs under actuator faults. However, it is still challenging to address the fault-tolerant formation control problem for multiple quadrotors in three aspects.

First, it is difficult to achieve the fault-tolerant formation for multiple quadrotors with nonlinearities and underactuation. In practical formation flight, each quadrotor in the team is a nonlinear system with six DOFs, but only four control inputs, which means that the independent control inputs are fewer than the degrees of freedom. The underactuated property of quadrotors will result in time-varying control input uncertainty for quadrotor position control subsystem. In [22], the quadrotor dynamics is simplified and the nonlinear dynamics in the quadrotor attitude dynamics was ignored. Furthermore, both the works in [22,23] ignored the effects of underactuation on the quadrotor team.

Second, the effects of different actuator faults on both the quadrotor position and attitude dynamics make it challenging to design fault-tolerant formation controller for multiple quadrotors. The different actuator faults in the vehicle dynamics of each quadrotor will result in a global heterogeneous multi-quadrotor system. both the works in [22,23] only focused on the fault-tolerant attitude control problem of quadrotors and did not consider the effects of multiple actuator faults on the quadrotor position dynamics in the fault-tolerant formation flight task.

Third, it is difficult to obtain a completely distributed formation controller for multiple quadrotors using only local information. In [23], the global information of leader dynamics was required in the distributed observer design of [23], which means

that the synchronization controller in [23] was not completely distributed.

Motivated by the above discussions, the effects of nonlinear dynamics, unknown system parameters, underactuation, and multiple actuator faults on both the position and attitude dynamics are considered in this paper and a completely distributed fault-tolerant formation controller is designed to achieve the fault-tolerant formation flight for multiple quadrotors with six DOFs. The designed formation controller includes a distributed observer to generate the position references, a fault-tolerant position control law to track the position references and generate the attitude references, and a fault-tolerant attitude control law to track the attitude references. The RL algorithms are designed for the quadrotor position and attitude subsystem to obtain the optimal control policies. Compared to previous works, the contributions of this paper are as follows:

- The underactuated property of quadrotors is considered to be the cause of time-varying control input uncertainty for quadrotor position control subsystem and fault-tolerant controller is designed to restrain the effect of underactuation on the quadrotor team.
- the effects of different actuator faults on both the quadrotor position and attitude dynamics are considered for the multi-quadrotor systems with six DOFs and a fault-tolerant formation controller is developed to achieve the optimal formation flight and restrain the actuator faults.
- A completely distributed formation controller for multiple quadrotors is obtained by designing distributed observer using only local information. Compared with the previous work of [23], the leader dynamics in this paper is considered to be only available for its neighbors. By using the system data of each quadrotor and its observer, RL algorithms are designed to learn optimal control policies online without the requirement of unknown system parameters.

The rest parts of this paper are organized as follows. The basic concepts on graph theory, the quadrotor vehicle dynamic model, the actuator fault model, and the problem formulation are provided in Section 2. A fault-tolerant formation control law is designed and the stability of the proposed formation control law is discussed in Section 3. The simulation results of a multi-quadrotor systems with multiple actuator faults are given in Section 4 and the conclusion is provided in Section 5.

Notations. I_N be a unit matrix, $\lambda_i(M)$ the i th eigenvalue of the matrix M , $\lambda_R(M)$ a set of the real parts of the eigenvalues of M , $0_{k \times l}$ a $k \times l$ zero matrix, and $c_{a,b}$ an $a \times 1$ vector with 1 in the b th element and 0s elsewhere.

2. Preliminaries

2.1. Basic concepts on graph theory

Consider a group of N quadrotors and let $\Phi = \{1, 2, \dots, N\}$. In this paper, the communication interaction among quadrotor agents is described by a directed graph $G = (V, E, W)$, where $V = \{v_i\}$ ($i \in \Phi$) indicates a set of nodes, $E \subset V \times V$ a set of edges, and $W = [w_{ij}] \in \mathbb{R}^{N \times N}$ the connectivity matrix. The weight w_{ij} satisfies that $w_{ij} > 0$ if $(v_i, v_j) \in E$ and $w_{ij} = 0$ otherwise. Define $N_i = \{v_j | (v_i, v_j) \in E\}$ as the neighbor set of v_i and d_i the in-degree of v_i with $d_i = \sum_{j=1}^N w_{ij}$. Let $D = \text{diag}(d_i) \in \mathbb{R}^{N \times N}$ and define L as a graph Laplacian matrix with $L = D - W$. A directed path from v_i to v_j satisfies $\{(v_i, v_k), (v_k, v_n), \dots, (v_l, v_j)\}$. G has a spanning tree on condition that a path exists in G connecting all the nodes. In this paper, it is assumed that a spanning tree exists in the communication graph G . Define $\varsigma = \text{diag}(\beta_i) \in \mathbb{R}^{N \times N}$, where β_i is a constant for quadrotor i satisfying that $\beta_i > 0$ if quadrotor i has access to the virtual leader and $\beta_i = 0$ otherwise.

2.2. Quadrotor model

Define $\hat{E}^I = \{e_x^I, e_y^I, e_z^I\}$ as the inertial frame and $\hat{E}^B = \{e_x^B, e_y^B, e_z^B\}$ the body-fixed frame. Define $p_i = [p_{i,x} \ p_{i,y} \ p_{i,z}]^T \in \mathbb{R}^3$ as the position vector and $\Theta_i = [\phi_i \ \theta_i \ \psi_i]^T \in \mathbb{R}^3$ the Euler angle vector of quadrotor i . Then, from [24], one can have the quadrotor dynamic model as:

$$\begin{aligned} m_i \ddot{p}_i &= R_{fi} F_i, \\ J_{\Theta i} \ddot{\Theta}_i &= -C(\Theta_i, \dot{\Theta}_i) \dot{\Theta}_i + \tau_i, \end{aligned} \quad (1)$$

where $m_i \in \mathbb{R}$ is the mass of quadrotor i , $J_{\Theta i} = \text{diag}(J_{\phi i}, J_{\theta i}, J_{\psi i}) \in \mathbb{R}^{3 \times 3}$ a vehicle inertial matrix, $C(\Theta_i, \dot{\Theta}_i) \in \mathbb{R}^{3 \times 3}$ a nonlinear Coriolis term shown in [25], and $R_{fi} \in SO(3)$ a transformation matrix from $\hat{E}^B$ to $\hat{E}^I$ satisfying (see, [24]) (see R_{fi} given in Box 1): $F_i \in \mathbb{R}^3$ is the external force and $\tau_i \in \mathbb{R}^3$ the torque generated by the vehicle rotors satisfying $F_i = c_{3,3} k_{\omega i} \sum_{k=1}^4 \omega_{i,k}^2 - R_{fi}^T c_{3,3} m_i g$, $\tau_i = [\tau_{i,x} \ \tau_{i,y} \ \tau_{i,z}]^T$, $\tau_{i,x} = k_{ti} k_{\omega i} (\omega_{i,1}^2 - \omega_{i,3}^2)$, $\tau_{i,y} = k_{ti} k_{\omega i} (\omega_{i,2}^2 - \omega_{i,4}^2)$, and $\tau_{i,z} = l_{ti} \sum_{k=1}^4 (-1)^{k+1} \omega_{i,k}^2$, respectively, where g is the gravity constant, $\omega_{i,k}$ ($k = 1, 2, 3, 4$) is the rotational speed of the k th rotor of quadrotor i , and k_{ti} , $k_{\omega i}$, l_{ti} are positive constants for quadrotor i . Define $\bar{\omega}_i^2 = [\omega_{i,1}^2 \ \omega_{i,2}^2 \ \omega_{i,3}^2 \ \omega_{i,4}^2]^T$. In practical applications, the vehicle control inputs are distributed as

$$\begin{bmatrix} u_{\phi i} \\ u_{\theta i} \\ u_{\psi i} \\ u_{zi} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \omega_{i,1}^2 \\ \omega_{i,2}^2 \\ \omega_{i,3}^2 \\ \omega_{i,4}^2 \end{bmatrix} = G_m \bar{\omega}_i^2. \quad (2)$$

Let $u_{pi} = [u_{pi,x} \ u_{pi,y} \ u_{pi,z}]^T \in \mathbb{R}^3$ be a virtual control input for the quadrotor position subsystem with

$$u_{pi} = R_{fi} c_{3,3} u_{zi}. \quad (3)$$

Then, the quadrotor vehicle model in (1) is rewritten as the following

$$\begin{aligned} \ddot{p}_i &= b_{pi} u_{pi} - c_{3,3} g, \\ \ddot{\Theta}_i &= -J_{\Theta i}^{-1} (C(\Theta_i, \dot{\Theta}_i) \dot{\Theta}_i) + b_{ti} u_{ti}, \end{aligned} \quad (4)$$

where $u_{ti} = [u_{\phi i} \ u_{\theta i} \ u_{\psi i}]^T \in \mathbb{R}^3$, $b_{pi} = m_i^{-1} k_{\omega i} I_{3 \times 3}$, and $b_{ti} = J_{\Theta i}^{-1} \text{diag}\{k_{ti} k_{\omega i}, k_{ti} k_{\omega i}, l_{ti}\}$. The quadrotor control system (4) can be seen as a cascade outer-inner control scheme including an outer control loop with the virtual control input u_{pi} and an inner control loop with the control input u_{ti} .

Remark 1. In practical applications, the system parameters b_{pi} and b_{ti} in (4) are unknown due to the unknown payloads that the quadrotor team carries in each formation flight. In this paper, an optimal control policy is learned using RL method without knowledge of the system parameters.

2.3. Actuator fault model

In practical formation flight of multiple quadrotor vehicles, the loss of effectiveness (LOE) in the actuator is a common actuator fault resulted from the structure damage to the propellers or the accidental variation in the parameters of rotors. Similarly to [26,27], the actuator fault model in the j th rotor of quadrotor i can be captured as $(\omega_{i,j}^*)^2 = \gamma_{ij} \omega_{i,j}^2$, $i \in \Phi$, $j = 1, 2, 3, 4$, where $\omega_{i,j}^*$ is the rotational speed in practical applications and γ_{ij} an unknown scaling factor with $0 < \gamma_{ij} \leq 1$. Consider the multiple actuator faults of the rotors and let $K_{\gamma i} = \text{diag}\{\gamma_{i1}, \gamma_{i2}, \gamma_{i3}, \gamma_{i4}\}$ ($i \in \Phi$). Then, the quadrotor dynamic model with multiple actuator faults can be written as

$$\begin{aligned} \ddot{p}_i &= b_{pi} (u_{pi} - u_{\Delta pi}) - c_{3,3} g, \quad i \in \Phi, \\ \ddot{\Theta}_i &= -J_{\Theta i}^{-1} (C(\Theta_i, \dot{\Theta}_i) \dot{\Theta}_i) + b_{ti} (u_{ti} - \mu(t - t_{si}) u_{\Delta \Theta i}), \end{aligned} \quad (5)$$

where $\mu(t - t_{si})$ is a function with an unknown fault starting time t_{si} for quadrotor i satisfying that $\mu(t - t_{si}) = 0$ if $t < t_{si}$ and otherwise, $\mu(t - t_{si}) = 1$. $u_{\Delta pi} \in \mathbb{R}^3$ is the position control input uncertainty resulted from the actuator faults and the underactuated property of the quadrotor team satisfying $u_{\Delta pi} = u_{pi} - u_{\Delta ai}$, $u_{\Delta ai} = R_{fi} c_{3,3} [u_{zi} - \mu(t - t_{si}) u_{\Delta zi}]$, and $u_{\Delta zi} = [1 - \gamma_{i1} \ 1 - \gamma_{i2} \ 1 - \gamma_{i3} \ 1 - \gamma_{i4}] \bar{\omega}_i^2$. $u_{\Delta \Theta i} \in \mathbb{R}^3$ in (5) indicates the attitude control input uncertainty resulted from the unknown multiple actuator faults satisfying $u_{\Delta \Theta i} = [I_3 \ 0_{3 \times 1}] G_m (I_4 - K_{\gamma i}) \bar{\omega}_i^2$.

Remark 2. In the practical formation application of multiple quadrotors, the rotational speed $\bar{\omega}_i^2$ of the quadrotor i remains bounded but varies all the time. Then, one can have that $u_{\Delta pi}$ and $u_{\Delta \Theta i}$ are time-varying and bounded with $\|u_{\Delta pi}\| \leq \bar{u}_{\Delta pi}$ and $\|u_{\Delta \Theta i}\| \leq \bar{u}_{\Delta \Theta i}$. The bounds $\bar{u}_{\Delta pi}$ and $\bar{u}_{\Delta \Theta i}$ are unknown in practical formation flight and are estimated by using fault observers.

2.4. Problem formulation

Define $p_0 = [p_{0,x} \ p_{0,y} \ p_{0,z}]^T \in \mathbb{R}^3$ as the position vector of the virtual leader in E^I and $\delta_i = [\delta_{i,x} \ \delta_{i,y} \ \delta_{i,z}]^T \in \mathbb{R}^3$ ($i \in \Phi$) the desired relative position between quadrotor i and the virtual leader. Let $\delta_{ij} = [\delta_{ij,x} \ \delta_{ij,y} \ \delta_{ij,z}]^T \in \mathbb{R}^3$ ($i, j \in \Phi$) be the predesigned relative position between quadrotor i and j with $\delta_{ij} = \delta_i - \delta_j$. Define $p_{ri} = [p_{ri,x} \ p_{ri,y} \ p_{ri,z}]^T \in \mathbb{R}^3$ as the position reference vector for quadrotor i satisfying $p_{ri} = p_0 + \delta_i$. For the distributed multi-quadrotor systems, it is considered that quadrotor i can only obtain state information from its neighbors and itself, which means that the leader dynamic information, the relative position between quadrotor i and the leader δ_i , and the position reference p_{ri} are unavailable and should be estimated for quadrotor i if quadrotor i is not connected to the leader. The main goal of this paper is to estimate the position reference for each quadrotor using distributed observer and then design fault-tolerant controllers to achieve the formation flight by tracking the estimated position reference.

3. Control protocol design

In this section, a data-driven fault-tolerant formation protocol including a distributed observer, a fault-tolerant position control law, and a fault-tolerant attitude control law is proposed to achieve the desired formation of cooperative quadrotor team.

3.1. Distributed observer design

Let $x_{p0} = [p_0^T \ \dot{p}_0^T]^T \in \mathbb{R}^6$ be the state vector of the virtual leader and the leader dynamics can be given as

$$\dot{x}_{p0} = A_{p0} x_{p0}, \quad (6)$$

where $A_{p0} \in \mathbb{R}^{6 \times 6}$ is the leader dynamic matrix and should be estimated for quadrotor i if quadrotor i has no access to the leader. Let $r_{pi} = [p_{ri} \ \dot{p}_{ri}]^T \in \mathbb{R}^6$ be the reference vector of quadrotor i and $\hat{r}_{pi} = [\hat{p}_{ri} \ \hat{\dot{p}}_{ri}]^T \in \mathbb{R}^6$ the estimated value of r_{pi} for quadrotor i . Let $\bar{\delta}_{ij} = [\delta_{ij}^T \ 0_{1 \times 3}]^T \in \mathbb{R}^6$ and $\bar{\delta}_i = [\delta_i^T \ 0_{1 \times 3}]^T \in \mathbb{R}^6$. In the following, a distributed observer is constructed as

$$\begin{aligned} \dot{\hat{A}}_{pi} &= \chi \varepsilon_{ai}, \quad \dot{\hat{\delta}}_i = \chi \varepsilon_{\delta i}, \\ \dot{\hat{r}}_{pi} &= \hat{A}_{pi} (\hat{r}_{pi} - \hat{\delta}_i) + \chi \varepsilon_{pi}, \end{aligned} \quad (7)$$

where $\hat{A}_{pi}$ is the estimated value of A_{pi} , χ a positive scaling factor to be determined, $\varepsilon_{ai} = \sum_{j \in N_i} (w_{ij} (\hat{A}_{pj} - \hat{A}_{pi}) + \beta_i (A_{p0} - \hat{A}_{pi}))$

$$R_{fi} = \begin{bmatrix} \cos \theta_i \cos \psi_i & \cos \psi_i \sin \theta_i \sin \phi_i - \cos \phi_i \sin \psi_i & \sin \phi_i \sin \psi_i + \cos \phi_i \cos \psi_i \sin \theta_i \\ \cos \theta_i \sin \psi_i & \cos \phi_i \cos \psi_i + \sin \phi_i \sin \psi_i \sin \theta_i & \cos \phi_i \sin \psi_i \sin \theta_i - \cos \psi_i \sin \psi_i \\ -\sin \theta_i & \cos \theta_i \sin \phi_i & \cos \phi_i \cos \theta_i \end{bmatrix}.$$

Box 1.

the local dynamic information estimation error, $\hat{\delta}_i$ the estimated value of δ_i , $\varepsilon_{\delta i} = \sum_{j \in N_i} (w_{ij} (\hat{\delta}_j - \hat{\delta}_i) + \beta_i (\bar{\delta}_i - \hat{\delta}_i))$ the local relative position estimation error, and $\varepsilon_{pi} = \sum_{j \in N_i} (w_{ij} (\hat{r}_{pj} - \hat{r}_{pi} + \bar{\delta}_{ij}) + \beta_i (x_{p0} + \bar{\delta}_i - \hat{r}_{pi}))$ the local position reference estimation error. Let $\tilde{r}_{pi} \in \mathbb{R}^6$ be the estimation error of the i th observer with

$$\tilde{r}_{pi} = \hat{r}_{pi} - \bar{\delta}_i - x_{p0}. \quad (8)$$

Theorem 1. Given the leader system in (6) and the distributed observer in (7), for any initial estimation errors $\tilde{A}_{pi}(0)$, $\tilde{\delta}_i(0)$, and $\tilde{r}_{pi}(0)$, the convergence of the estimation errors will be achieved, i.e., $\lim_{t \rightarrow \infty} \tilde{A}_{pi}(t) = 0$, $\lim_{t \rightarrow \infty} \tilde{\delta}_i(t) = 0$, and $\lim_{t \rightarrow \infty} \tilde{r}_{pi}(t) = 0$, if χ is sufficiently large.

Proof. Define $\tilde{\tilde{A}}_p = [\tilde{\tilde{A}}_{p1}^T \quad \tilde{\tilde{A}}_{p2}^T \quad \dots \quad \tilde{\tilde{A}}_{pN}^T]^T$. Then, one can have from (7) that

$$\dot{\tilde{\tilde{A}}}_p = -\chi [(L + \varsigma) \otimes I_6] \tilde{\tilde{A}}_p. \quad (9)$$

Since a spanning tree exists in the graph G , one can obtain from Lemma 4 of [28] that all the real parts of the eigenvalues of the matrix $L + \varsigma$ are positive. Therefore, one can have $\lim_{t \rightarrow \infty} \tilde{\tilde{A}}_p(t) = 0$ for any $\chi > 0$, which implies that for quadrotor i ($i \in \Phi$), $\lim_{t \rightarrow \infty} \tilde{\tilde{A}}_{pi}(t) = 0$. Define $\tilde{\tilde{\delta}} = [\tilde{\tilde{\delta}}_1^T \quad \tilde{\tilde{\delta}}_2^T \quad \dots \quad \tilde{\tilde{\delta}}_N^T]^T$. Similarly, one can from (7) that $\dot{\tilde{\tilde{\delta}}} = -\chi [(L + \varsigma) \otimes I_6] \tilde{\tilde{\delta}}$, which yields that $\lim_{t \rightarrow \infty} \tilde{\tilde{\delta}}_i(t) = 0$.

Then, differentiating (8) and using (6) and (7) lead to

$$\begin{aligned} \dot{\tilde{r}}_{pi} &= \dot{\hat{r}}_{pi} - \dot{x}_{p0} = \hat{A}_{pi} (\hat{r}_{pi} - \hat{\delta}_i) - A_{p0} (\hat{r}_{pi} - \hat{\delta}_i) \\ &+ A_{p0} (\hat{r}_{pi} - \hat{\delta}_i) + \chi \varepsilon_{pi} - A_{p0} x_{p0} \\ &= \tilde{A}_{pi} (\hat{r}_{pi} - \hat{\delta}_i) + \chi \varepsilon_{pi} + A_{p0} (\hat{r}_{pi} - \hat{\delta}_i - x_{p0}) \\ &= \tilde{A}_{pi} (\hat{r}_{pi} - \bar{\delta}_i - \tilde{\delta}_i) + \chi \varepsilon_{pi} + A_{p0} (\hat{r}_{pi} - \bar{\delta}_i - \tilde{\delta}_i - x_{p0}) \\ &= \tilde{A}_{pi} (x_{p0} - \tilde{\delta}_i) + \tilde{A}_{pi} \tilde{r}_{pi} + \chi \varepsilon_{pi} + A_{p0} \tilde{r}_{pi} - A_{p0} \tilde{\delta}_i \end{aligned} \quad (10)$$

From (10), one can obtain that

$$\dot{\tilde{r}}_p = (\tilde{A}_0 + \tilde{A}_p) \tilde{r}_p + \tilde{A}_p (1_N \otimes x_{p0}) - \tilde{A}_p \tilde{\delta} - (I_N \otimes A_{p0}) \tilde{\delta}, \quad (11)$$

where $\tilde{r}_p = [\tilde{r}_{p1}^T \quad \tilde{r}_{p2}^T \quad \dots \quad \tilde{r}_{pN}^T]^T$, $\tilde{A}_p = \text{diag}(\tilde{A}_{p1}, \tilde{A}_{p2}, \dots, \tilde{A}_{pN})$, and $\tilde{A}_0 = I_N \otimes A_{p0} - \chi (L + \varsigma) \otimes I_6$. Note that $\tilde{\tilde{A}}_p$ and $\tilde{\tilde{\delta}}$ will converge to 0s if $\chi > 0$. Then, one can have that $\tilde{r}_p$ will converge to 0 if $\tilde{A}_0$ is Hurwitz. The eigenvalues of $\tilde{A}_0$ in (11) can be obtained as $\{\lambda_j(A_{p0}) - \chi \lambda_i(L + \varsigma) | j = 1, 2, \dots, 6, i \in \Phi\}$. Since a spanning tree exists in the graph G , one can obtain that all the real parts of the eigenvalues of the matrix $L + \varsigma$ are positive. Then, $\tilde{A}_0$ is Hurwitz if $\chi > \max(\lambda_R(A_{p0})) / \min(\lambda_R(L + \varsigma))$, which implies that $\lim_{t \rightarrow \infty} \tilde{r}_{pi}(t) = 0$, if χ is sufficiently large.

Remark 3. By using the distributed observer in (7), the complicated formation mission can be decomposed as trajectory tracking mission for each quadrotor. In fact, the formation flight can

be achieved for each quadrotor by tracking the states of its observer, independently. Therefore, the position controllers for each quadrotor involve only the states of its observer and itself and are low in state dimension, compared to the controller designs of [5,24].

3.2. Position controller design

In this subsection, a data-driven position FTC law is designed to achieve the desired position formation for multiple quadrotors under multiple actuator faults in the quadrotor rotors. From (5) and (7), a position augmented subsystem can be given as

$$\dot{X}_{pi} = \bar{A}_{pi} X_{pi} + \bar{B}_{pi} (u_{pi} - u_{\Delta pi}) - c_{12,6}g + D_{pi} \varepsilon_{pi}, \quad (12)$$

where $X_{pi} = [x_{pi} \quad \hat{r}_{pi}]^T \in \mathbb{R}^{12}$, $\bar{A}_{pi} = \text{diag}(A_{pi}, A_{p0}) \in \mathbb{R}^{12 \times 12}$, $\bar{B}_{pi} = [B_{pi}^T \quad 0_{3 \times 6}]^T \in \mathbb{R}^{12 \times 3}$, $B_{pi} = [0_{3 \times 3} \quad b_{pi}^T]^T \in \mathbb{R}^{6 \times 3}$, and $\bar{D}_{pi} = [0_{6 \times 6} \quad \chi I_6]^T \in \mathbb{R}^{12 \times 6}$. According to Theorem 1, $\hat{r}_{pi}$ will converge to r_{pi} and thus ε_{pi} will converge to 0. Construct the position tracking performance function as

$$V_{pi}(e_{pi}, u_{pi}) = \int_t^\infty e^{-\alpha_{1i}(\tau-t)} (e_{pi}^T Q_{pi} e_{pi} + u_{pi}^T R_{pi} u_{pi}) d\tau, \quad (13)$$

where $e_{pi} = [e_{pi,x} \quad e_{pi,y} \quad e_{pi,z}]^T \in \mathbb{R}^3$ is a position tracking error vector for quadrotor i satisfying $e_{pi} = p_i - \hat{p}_{pi}$, $Q_{pi} > 0$, $R_{pi} > 0$, and $\alpha_{1i} \geq 0$. By differentiating (13), one can obtain that

$$\begin{aligned} &e_{pi}^T Q_{pi} e_{pi} + u_{pi}^T R_{pi} u_{pi} - \alpha_{1i} V_{pi} \\ &+ \Delta V_{pi} (\bar{A}_{pi} X_{pi} + \bar{B}_{pi} u_{pi} - c_{12,6}g + D_{pi} \varepsilon_{pi}) = 0, \end{aligned}$$

where $\Delta V_{pi} = \partial V_{pi} / \partial X_{pi}$. Based on the traditional optimal control theory (see, e.g., [29]), the optimal control policy u_{pi}^* can be obtained as

$$u_{pi}^* = -\frac{1}{2} R_{pi}^{-1} \bar{B}_{pi}^T \Delta V_{pi}^*, \quad (14)$$

where V_{pi}^* is the solution to the following Hamilton–Jacobi–Bellman (HJB) equation:

$$\begin{aligned} &e_{pi}^T Q_{pi} e_{pi} - \frac{1}{4} (\Delta V_{pi}^*)^T \bar{B}_{pi} R_{pi}^{-1} \bar{B}_{pi}^T \Delta V_{pi}^* - \alpha_{1i} V_{pi} \\ &+ (\Delta V_{pi}^*)^T (\bar{A}_{pi} X_{pi} - c_{12,6}g + D_{pi} \varepsilon_{pi}) = 0. \end{aligned} \quad (15)$$

Considering the effects of multiple actuator faults on the quadrotor position dynamics, an adaptive position FTC law is designed as

$$u_{pi}(t) = \frac{1}{2} u_{pi}^*(x) + \frac{\hat{u}_{\Delta pi}^2 R_{pi} u_{pi}^*(x)}{\|R_{pi} u_{pi}^*(x)\| \hat{u}_{\Delta pi} + \sigma_1(t)}, \quad (16)$$

where $\hat{u}_{\Delta pi}$ is the estimated bound of the position control input uncertainty resulted from the actuator faults and $\sigma_1(t)$ a positive function with $\int_t^\infty \sigma_1(t) d\tau \leq \bar{\sigma}_1 < \infty$. The fault observer is used to estimate $\bar{u}_{\Delta pi}$ with

$$d\hat{u}_{\Delta pi}/dt = -k_{\sigma 1} \sigma_1(t) \hat{u}_{\Delta pi} + 2k_{\sigma 1} \|R_{pi} u_{pi}^*(x)\|, \quad (17)$$

where $k_{\sigma 1}$ is a positive scaling factor.

Remark 4. It can be observed from (16) that the proposed adaptive FTC law is dependent on the optimal policy u_{pi}^* . However, the traditional optimal control approach in (14), (15) relies on the quadrotor position dynamic model, which is difficult to obtain in practical formation test.

Therefore, in the following, a data-driven RL position control algorithm is designed to obtain u_{pi}^* without requirement of the vehicle dynamics. The augmented position control subsystem (12) is controlled by a stabilizing control policy $u_{pi}^0 \in \mathbb{R}^3$ and explored by a bounded signal $u_{pei} \in \mathbb{R}^3$ at the time t with $t < t_{si}$:

$$\begin{aligned} \dot{X}_{pi} &= \bar{A}_{pi}X_{pi} + \bar{B}_{pi}u_{pi}^n + \bar{B}_{pi}(u_{pi}^0 + u_{pei} - u_{pi}^n) - c_{12,6}g + D_{pi}\varepsilon_{pi}, \\ i &\in \Phi, \end{aligned} \quad (18)$$

where u_{pi}^n is the updated control policy in the n th iteration. Then, from the mathematical derivatives of (18), one can have the following Bellman equation as:

$$\begin{aligned} &e^{-\alpha_{1i}\Delta t} V_{pi}^n(X_{pi}(t + \Delta t)) - V_{pi}^n(X_{pi}(t)) \\ &= \int_t^{t+\Delta t} e^{-\alpha_{1i}(\tau-t)} \left[-e_{pi}^T Q_{pi} e_{pi} - (u_{pi}^n)^T R_{pi} u_{pi}^n \right] d\tau \\ &\quad - \int_t^{t+\Delta t} 2e^{-\alpha_{1i}(\tau-t)} (u_{pi}^{n+1})^T R_{pi} (u_{pi}^0 + u_{pei} - u_{pi}^n) d\tau \\ &\quad + \int_t^{t+\Delta t} 2e^{-\alpha_{1i}(\tau-t)} \Delta V_{pi}^n D_{pi} \varepsilon_{pi} d\tau, \end{aligned} \quad (19)$$

where $V_{pi}^n(X_{pi})$ is the updated performance function in the n th iteration. It can be observed that V_{pi}^n and u_{pi}^{n+1} can be updated simultaneously using (19), which leads to the RL algorithm in Algorithm 1. One can see from Algorithm 1 that the Bellman equation in (19) is updated iteratively until the convergence criterion is achieved.

The policies V_{pi}^n and u_{pi}^{n+1} in Algorithm 1 can be approximated by a critic neural network (NN) and an actor NN as follows:

$$\begin{aligned} \hat{V}_{pi}^n(X_{pi}) &= W_{p1i} \varphi_{p1i}(X_{pi}), \\ \hat{u}_{pi}^{n+1}(X_{pi}) &= W_{p2i} \varphi_{p2i}(X_{pi}), i \in \Phi, \end{aligned} \quad (20)$$

where $\hat{V}_{pi}^n$ and $\hat{u}_{pi}^{n+1}$ are the approximated values of V_{pi}^n and u_{pi}^{n+1} , $W_{p1i} \in \mathbb{R}^{1 \times l_{p1}}$, $W_{p2i} \in \mathbb{R}^{3 \times l_{p2}}$ the weights, l_{p1} , l_{p2} the numbers of neurons and $\varphi_{p1i} \in \mathbb{R}^{l_{p1} \times 1}$, $\varphi_{p2i} \in \mathbb{R}^{l_{p2} \times 1}$ the base functions. W_{1i} and W_{2i} in (20) can be updated using the least-square (LS) approach under a persistence excitation (PE) condition, as shown in [29]. Let $\Theta_{ri} = [\phi_{ri} \ \theta_{ri} \ \psi_{ri}]^T$ be the attitude reference vector.

Algorithm 1. Data-driven RL algorithm for position controller

1. Initialization:

Implement a fixed position control input u_{pi} with a bounded exploring signal u_{pei} to the quadrotor vehicle and collect the vehicle system data. Determine the stopping criterion $\vartheta_{11} > 0$.

2. Policy evaluation:

Solve Eq. (19) for V_{pi}^n and u_{pi}^{n+1} , simultaneously.

3. Policy improvement:

Let $u_{pi}^n = u_{pi}^{n+1}$ and stop if: $\|u_{pi}^{n+1} - u_{pi}^n\| < \vartheta_{11}$, otherwise, let $n = n + 1$ and go to Step 1.

After the control input u_{pi} in (16) is determined, one can obtain ϕ_{ri} , θ_{ri} , and u_{zi} by solving (3) for the attitude controller with

$$\begin{aligned} \phi_{ri} &= \arcsin((\cos \phi_i \sin \theta_i \sin \psi_i - u_{pyi}/u_{zi})/\cos \psi_i), \\ \theta_{ri} &= \arcsin((u_{pxi}/u_{zi} - \sin \phi_i \sin \psi_i)/\cos \psi_i/\cos \phi_i), \\ u_{zi} &= u_{pzi}/\cos \theta_i/\cos \phi_i. \end{aligned}$$

In practical flight, ψ_{ri} for the quadrotor i is preset as 0 in general. An attitude control law will be introduced in the following to achieve the attitude references tracking task.

3.3. Attitude controller design

Define $x_{\Theta ri} = [\Theta_{ri}^T \ \dot{\Theta}_{ri}^T]^T \in \mathbb{R}^6$ as the state vector of the attitude reference. The dynamics of the attitude reference is described as

$$\dot{x}_{\Theta ri} = F_{\Theta ri}(x_{\Theta ri}), i \in \Phi, \quad (21)$$

where $F_{\Theta ri}(x_{\Theta ri}) \in \mathbb{R}^6$ is an unknown nonlinear function. From (5), the vehicle attitude dynamic model can be rewritten as

$$\dot{x}_{\Theta i} = F_{\Theta i}(x_{\Theta i}) + B_{\tau i}(u_{\tau i} - \mu(t - t_{si})u_{\Delta \Theta i}), i \in \Phi, \quad (22)$$

where $x_{\Theta i} = [\Theta_i^T \ \dot{\Theta}_i^T]^T \in \mathbb{R}^6$ is the attitude state vector, $B_{\tau i} = [c_{6,4}b_{\tau i,1} \ c_{6,5}b_{\tau i,2} \ c_{6,6}b_{\tau i,3}]$, and $F_{\Theta i}(x_{\Theta i}) \in \mathbb{R}^6$ is a nonlinear function satisfying

$$F_{\Theta i}(x_{\Theta i}) = \begin{bmatrix} 0_{3 \times 3} & I_3 \\ 0_{3 \times 3} & -J_i^{-1}C(\Theta_i, \dot{\Theta}_i) \end{bmatrix} x_{\Theta i}.$$

Then, from (21), (22), one can construct an augmented attitude subsystem as

$$\dot{X}_{\Theta i} = \bar{F}_{\Theta}(X_{\Theta i}) + \bar{B}_{\tau i}(u_{\tau i} - \mu(t - t_{si})u_{\Delta \Theta i}), \quad (23)$$

where $X_{\Theta i} = [x_{\Theta i}^T \ x_{\Theta ri}^T]^T \in \mathbb{R}^{12}$, $\bar{F}_{\Theta}(X_{\Theta i}) = [F_{\Theta i}(x_{\Theta i})^T \ F_{\Theta ri}(x_{\Theta ri})^T]^T \in \mathbb{R}^{12}$, and $\bar{B}_{\tau i} = [B_{\tau i}^T \ 0]^T$. Let $e_{\Theta i} = [e_{\phi i} \ e_{\theta i} \ e_{\psi i}]^T \in \mathbb{R}^3$ be the attitude tracking error vector of quadrotor i with $e_{\Theta i} = \Theta_i - \Theta_i^r$. Similarly to the position controller design, one can define an attitude tracking performance function as

$$V_{\Theta i}(e_{\Theta i}, u_{\tau i}) = \int_t^\infty e^{-\alpha_{2i}(\tau-t)} (e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} + u_{\tau i}^T R_{\Theta i} u_{\tau i}) d\tau, \quad (24)$$

where $\alpha_{2i} \geq 0$, $Q_{\Theta i} = Q_{\Theta i}^T > 0$, and $R_{\Theta i} = R_{\Theta i}^T > 0$. Then, one can have an optimal attitude control policy $u_{\tau i}^*$ as

$$u_{\tau i}^* = -\frac{1}{2} R_{\Theta i}^{-1} \bar{B}_{\tau i}^T \Delta V_{\Theta i}^*, \quad (25)$$

where $V_{\Theta i}^*$ is an optimal attitude tracking performance function and satisfies the following HJB equation:

$$\begin{aligned} &e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} - \frac{1}{4} (\Delta V_{\Theta i}^*)^T \bar{B}_{\tau i} R_{\Theta i}^{-1} \bar{B}_{\tau i}^T \Delta V_{\Theta i}^* - \alpha_{2i} V_{\Theta i} \\ &\quad + (\Delta V_{\Theta i}^*)^T \bar{F}_{\Theta}(X_{\Theta i}) = 0. \end{aligned} \quad (26)$$

One can see from (26) that the nonlinear vehicle dynamic model is required in the traditional optimal controller design, which is unpractical in practical formation flight. In the following, a data-driven RL algorithm is given to iteratively learn the optimal policy $u_{\tau i}^*$ as Algorithm 2.

Algorithm 2. Data-driven RL algorithm for attitude controller

1. Initialization:

Start with a stabilizing control input $u_{\tau i}^0$ with a bounded exploring signal $u_{\tau ei}$ to the vehicle system and collect the system data. Determine the stopping criterion $\vartheta_{12} > 0$.

2. Policy evaluation:

Solve for $V_{\theta i}^n$ and $u_{\tau i}^{n+1}$, simultaneously:

$$e^{-\alpha_{2i}\Delta t} V_{\theta i}^n(X_{\theta i}(t+T)) - V_{\theta i}^n(X_{\theta i}(t)) = \int_t^{t+\Delta t} e^{-\alpha_{2i}(\tau-t)} \left(-e_{\theta i}^T Q_{\theta i} e_{\theta i} - (u_{\tau i}^n)^T R_{\theta i} u_{\tau i}^n \right) d\tau - \int_t^{t+\Delta t} e^{-\alpha_{2i}(\tau-t)} 2(u_{\tau i}^{n+1})^T R_{\theta i} (u_{\tau i}^0 + u_{\tau ei} - u_{\tau i}^n) d\tau. \quad (27)$$

3. Policy improvement:

Let $u_{\tau i}^n = u_{\tau i}^{n+1}$ and stop if: $\|u_{\tau i}^{n+1} - u_{\tau i}^n\| < \vartheta_{l2}$, otherwise, let $n = n + 1$ and go to Step 1.

For each quadrotor vehicle in the multi-quadrotor system, two NNs can be implemented to approximate the performance function $V_{\theta i}^n$ and the control protocol $u_{\tau i}^{n+1}$ with $\hat{V}_{\theta i}^n(X_{\theta i}) = \hat{W}_{\theta 1i} \varphi_{\theta 1i}(X_{\theta i})$, $\hat{u}_{\tau i}^{n+1}(X_{\theta i}) = \hat{W}_{\theta 2i} \varphi_{\theta 2i}(X_{\theta i})$, where $\hat{V}_{\theta i}^n(X_{\theta i})$, $\hat{u}_{\tau i}^{n+1}(X_{\theta i})$ are the approximated policies, $\hat{W}_{\theta 1i} \in \mathbb{R}^{l_{\theta 1}}$, $\hat{W}_{\theta 2i} \in \mathbb{R}^{3 \times l_{\theta 2}}$ the weights, $\varphi_{\theta 1i}(X_{\theta i}) \in \mathbb{R}^{l_{\theta 1}}$, $\varphi_{\theta 2i}(X_{\theta i}) \in \mathbb{R}^{l_{\theta 2}}$ the activation functions, and $l_{\theta 1}$, $l_{\theta 2}$ the numbers of neurons. Similarly to Algorithm 1, the weights $\hat{W}_{\theta 1i}$, $\hat{W}_{\theta 2i}$ can be iteratively updated by using the LS approach under the PE condition.

By using the learned attitude control policy, an adaptive FTC law is designed to address the multiple actuator faults for the quadrotor attitude subsystem as

$$u_{\tau i}(t) = \frac{1}{2} u_{\tau i}^*(x) + \frac{\hat{u}_{\Delta \theta i}^2 R_{\theta i} u_{\tau i}^*(x)}{\|R_{\theta i} u_{\tau i}^*(x)\| \hat{u}_{\Delta \theta i} + \sigma_2(t)}, \quad (28)$$

where $\sigma_2(t)$ is a positive function with $\int_t^\infty \sigma_2(\tau) d\tau \leq \bar{\sigma}_2 < \infty$. The bound of the attitude control uncertainty $\hat{u}_{\Delta \theta i}$ is estimated by a fault observer as

$$d\hat{u}_{\Delta \theta i}/dt = -k_{\sigma 2} \sigma_2(t) \hat{u}_{\Delta \theta i} + 2k_{\sigma 2} \|R_{\theta i} u_{\tau i}^*(x)\|, \quad (29)$$

where $\hat{u}_{\Delta \theta i}$ indicates the estimated value $\bar{u}_{\Delta \theta i}$ and $k_{\sigma 2}$ a positive constant.

3.4. Stability analysis

Define $\tilde{u}_{\Delta pi}$ and $\tilde{u}_{\Delta \theta i}$ as the estimation errors of the input uncertainties with $\tilde{u}_{\Delta pi} = \hat{u}_{\Delta pi} - \bar{u}_{\Delta pi}$ and $\tilde{u}_{\Delta \theta i} = \hat{u}_{\Delta \theta i} - \bar{u}_{\Delta \theta i}$. Form (17), (29), one can obtain the uncertainty estimation error dynamics as

$$\begin{aligned} d\tilde{u}_{\Delta pi}/dt &= -k_{\sigma 1} \sigma_1(t) (\tilde{u}_{\Delta pi} + \bar{u}_{\Delta pi}) + 2k_{\sigma 1} \|R_{pi} u_{pi}^*(x)\|, \\ d\tilde{u}_{\Delta \theta i}/dt &= -k_{\sigma 2} \sigma_2(t) (\tilde{u}_{\Delta \theta i} + \bar{u}_{\Delta \theta i}) + 2k_{\sigma 2} \|R_{\theta i} u_{\tau i}^*(x)\|. \end{aligned} \quad (30)$$

Theorem 2. Consider the multi-agent quadrotor team (5) controlled by the fault-tolerant position controller (16) and the fault-tolerant attitude controller (28), the position and attitude tracking errors e_{pi} and $e_{\theta i}$ will converge to 0s.

Proof. Design the following Lyapunov function as

$$L(V_{pi}^*, V_{\theta i}^*, \tilde{u}_{\Delta pi}, \tilde{u}_{\Delta \theta i}) = V_{pi}^* + V_{\theta i}^* + \frac{1}{2} k_{\sigma 1}^{-1} \tilde{u}_{\Delta pi}^2 + \frac{1}{2} k_{\sigma 2}^{-1} \tilde{u}_{\Delta \theta i}^2. \quad (31)$$

Differentiating (31) and using the position and attitude augmented system dynamic models in (12), (23) yield

$$dL(V_{pi}^*, V_{\theta i}^*, \tilde{u}_{\Delta pi}, \tilde{u}_{\Delta \theta i})/dt = (\Delta V_{pi}^*)^T [\bar{A}_{pi} X_{pi} + \bar{B}_{pi} (u_{pi} - u_{\Delta pi}) - c_{12.6g}] + (\Delta V_{\theta i}^*)^T \bar{F}_{\theta}(X_{\theta i})$$

$$+ (\Delta V_{\theta i}^*)^T \bar{B}_{\tau i} (u_{\tau i} - \mu(t - t_{si}) u_{\Delta \theta i}) + k_{\sigma 1}^{-1} \tilde{u}_{\Delta pi} \dot{\tilde{u}}_{\Delta pi} + k_{\sigma 1}^{-1} \tilde{u}_{\Delta \theta i} \dot{\tilde{u}}_{\Delta \theta i} + (\Delta V_{pi}^*)^T D_{pi} \varepsilon_{pi}. \quad (32)$$

By using the HJB equations in (15) and (26), one can obtain from (32) that

$$\begin{aligned} dL(V_{pi}^*, V_{\theta i}^*, \tilde{u}_{\Delta pi}, \tilde{u}_{\Delta \theta i})/dt &= -e_{pi}^T Q_{pi} e_{pi} + \frac{1}{4} (\Delta V_{pi}^*)^T \bar{B}_{pi} R_{pi}^{-1} \bar{B}_{pi}^T \Delta V_{pi}^* \\ &+ \alpha_{1i} V_{pi}^* + (\Delta V_{pi}^*)^T \bar{B}_{pi} (u_{pi} - u_{\Delta pi}) \\ &+ k_{\sigma 1}^{-1} \tilde{u}_{\Delta pi} \dot{\tilde{u}}_{\Delta pi} - e_{\theta i}^T Q_{\theta i} e_{\theta i} + \frac{1}{4} (\Delta V_{\theta i}^*)^T \bar{B}_{\tau i} R_{\tau i}^{-1} \bar{B}_{\tau i}^T \Delta V_{\theta i}^* + \alpha_{2i} V_{\theta i}^* \\ &+ (\Delta V_{\theta i}^*)^T \bar{B}_{\tau i} (u_{\tau i} - \mu(t - t_{si}) u_{\Delta \theta i}) + k_{\sigma 1}^{-1} \tilde{u}_{\Delta \theta i} \dot{\tilde{u}}_{\Delta \theta i}. \end{aligned} \quad (33)$$

Subscribing the designed FTC laws in (16) and (28) into (33) yields

$$\begin{aligned} dL(V_{pi}^*, V_{\theta i}^*, \tilde{u}_{\Delta pi}, \tilde{u}_{\Delta \theta i})/dt &= -e_{pi}^T Q_{pi} e_{pi} + \frac{1}{4} (\Delta V_{pi}^*)^T \bar{B}_{pi} R_{pi}^{-1} \bar{B}_{pi}^T \Delta V_{pi}^* + \alpha_{1i} V_{pi}^* - (\Delta V_{pi}^*)^T \bar{B}_{pi} u_{\Delta pi} \\ &+ k_{\sigma 1}^{-1} \tilde{u}_{\Delta pi} \dot{\tilde{u}}_{\Delta pi} + (\Delta V_{pi}^*)^T \bar{B}_{pi} \left(\frac{1}{2} u_{pi}^*(x) + \frac{\hat{u}_{\Delta pi}^2 R_{pi} u_{pi}^*(x)}{\|R_{pi} u_{pi}^*(x)\| \hat{u}_{\Delta pi} + \sigma_1(t)} \right) \\ &- e_{\theta i}^T Q_{\theta i} e_{\theta i} + \frac{1}{4} (\Delta V_{\theta i}^*)^T \bar{B}_{\tau i} R_{\tau i}^{-1} \bar{B}_{\tau i}^T \Delta V_{\theta i}^* \\ &+ \alpha_{2i} V_{\theta i}^* - (\Delta V_{\theta i}^*)^T \bar{B}_{\tau i} \mu(t - t_{si}) u_{\Delta \theta i} + k_{\sigma 2}^{-1} \tilde{u}_{\Delta \theta i} \dot{\tilde{u}}_{\Delta \theta i} \\ &+ (\Delta V_{\theta i}^*)^T \bar{B}_{\tau i} \left(\frac{1}{2} u_{\tau i}^*(x) + \frac{\hat{u}_{\Delta \theta i}^2 R_{\theta i} u_{\tau i}^*(x)}{\|R_{\theta i} u_{\tau i}^*(x)\| \hat{u}_{\Delta \theta i} + \sigma_2(t)} \right). \end{aligned} \quad (34)$$

By substituting the optimal control policies in (14), (25) and the uncertainty estimation error dynamics in (30) into (34), one can have

$$\begin{aligned} dL(V_{pi}^*, V_{\theta i}^*, \tilde{u}_{\Delta pi}, \tilde{u}_{\Delta \theta i})/dt &= -e_{pi}^T Q_{pi} e_{pi} + \alpha_{1i} V_{pi}^* - (\Delta V_{pi}^*)^T \bar{B}_{pi} u_{\Delta pi} - e_{\theta i}^T Q_{\theta i} e_{\theta i} + \alpha_{2i} V_{\theta i}^* \\ &- (\Delta V_{\theta i}^*)^T \bar{B}_{\tau i} \mu(t - t_{si}) u_{\Delta \theta i} - \frac{2\hat{u}_{\Delta pi}^2 \|R_{pi} u_{pi}^*\|^2}{\|R_{pi} u_{pi}^*\| \hat{u}_{\Delta pi} + \sigma_1(t)} \\ &- \frac{2\hat{u}_{\Delta \theta i}^2 \|R_{\theta i} u_{\tau i}^*\|^2}{\|R_{\theta i} u_{\tau i}^*\| \hat{u}_{\Delta \theta i} + \sigma_2(t)} \\ &+ \tilde{u}_{\Delta pi} (-\sigma_1(t) (\tilde{u}_{\Delta pi} + \bar{u}_{\Delta pi}) + 2 \|R_{pi} u_{pi}^*\|) \\ &+ \tilde{u}_{\Delta \theta i} (-\sigma_2(t) (\tilde{u}_{\Delta \theta i} + \bar{u}_{\Delta \theta i}) + 2 \|R_{\theta i} u_{\tau i}^*\|). \end{aligned} \quad (35)$$

Considering the fact that $u_{\Delta pi}(t) \leq \bar{u}_{\Delta pi}$ and $\mu(t - t_{si}) u_{\Delta \theta i} \leq \bar{u}_{\Delta \theta i}$, one can obtain from (35) that

$$\begin{aligned} dL(V_{pi}^*, V_{\theta i}^*, \tilde{u}_{\Delta pi}, \tilde{u}_{\Delta \theta i})/dt &\leq -e_{pi}^T Q_{pi} e_{pi} + \alpha_{1i} V_{pi}^* - \tilde{u}_{\Delta pi} \sigma_1(t) (\tilde{u}_{\Delta pi} + \bar{u}_{\Delta pi}) \\ &+ 2 \|R_{pi} u_{pi}^*\| (\tilde{u}_{\Delta pi} + \bar{u}_{\Delta pi}) \\ &- \frac{2\hat{u}_{\Delta pi}^2 \|R_{pi} u_{pi}^*\|^2}{\|R_{pi} u_{pi}^*\| \hat{u}_{\Delta pi} + \sigma_1(t)} - e_{\theta i}^T Q_{\theta i} e_{\theta i} + \alpha_{2i} V_{\theta i}^* \\ &- \tilde{u}_{\Delta \theta i} \sigma_2(t) (\tilde{u}_{\Delta \theta i} + \bar{u}_{\Delta \theta i}) \\ &+ 2 \|R_{\theta i} u_{\tau i}^*\| (\tilde{u}_{\Delta \theta i} + \bar{u}_{\Delta \theta i}) - \frac{2\hat{u}_{\Delta \theta i}^2 \|R_{\theta i} u_{\tau i}^*\|^2}{\|R_{\theta i} u_{\tau i}^*\| \hat{u}_{\Delta \theta i} + \sigma_2(t)} \end{aligned} \quad (36)$$

It should be noted that

$$\begin{aligned} 2 \|R_{pi} u_{pi}^*\| (\tilde{u}_{\Delta pi} + \tilde{u}_{\Delta pi}) - \frac{2\tilde{u}_{\Delta pi}^2 \|R_{pi} u_{pi}^*\|^2}{\|R_{pi} u_{pi}^*\| \|\tilde{u}_{\Delta pi} + \sigma_1(t)\|^2} &\leq 2\sigma_1(t), \\ 2 \|R_{\theta i} u_{\theta i}^*\| (\tilde{u}_{\Delta \theta i} + \tilde{u}_{\Delta \theta i}) - \frac{2\tilde{u}_{\Delta \theta i}^2 \|R_{\theta i} u_{\theta i}^*\|^2}{\|R_{\theta i} u_{\theta i}^*\| \|\tilde{u}_{\Delta \theta i} + \sigma_2(t)\|^2} &\leq 2\sigma_2(t). \end{aligned} \quad (37)$$

From (36), (37), one can have

$$\begin{aligned} dL(V_{pi}^*, V_{\theta i}^*, \tilde{u}_{\Delta pi}, \tilde{u}_{\Delta \theta i})/dt &\leq \\ &- e_{pi}^T Q_{pi} e_{pi} + \alpha_{1i} V_{pi}^* - e_{\theta i}^T Q_{\theta i} e_{\theta i} + \alpha_{2i} V_{\theta i}^* \\ &+ \frac{1}{4} \sigma_1(t) [8 + \tilde{u}_{\Delta pi}^2 - (2\tilde{u}_{\Delta pi} + \tilde{u}_{\Delta pi})^2] \\ &+ \frac{1}{4} \sigma_2(t) [8 + \tilde{u}_{\Delta \theta i}^2 - (2\tilde{u}_{\Delta \theta i} + \tilde{u}_{\Delta \theta i})^2] \\ &\leq -e_{pi}^T Q_{pi} e_{pi} + \alpha_{1i} V_{pi}^* - e_{\theta i}^T Q_{\theta i} e_{\theta i} + \alpha_{2i} V_{\theta i}^* + \frac{1}{4} \sigma_1(t) (8 + \tilde{u}_{\Delta pi}^2) \\ &+ \frac{1}{4} \sigma_2(t) (8 + \tilde{u}_{\Delta \theta i}^2). \end{aligned} \quad (38)$$

Select the scaling factors α_{1i} and α_{2i} with

$$\int_t^\infty \alpha_{1i}(\tau) d\tau \leq \bar{\alpha}_{1i} < \infty, \int_t^\infty \alpha_{2i}(\tau) d\tau \leq \bar{\alpha}_{2i} < \infty. \quad (39)$$

Because the performance functions V_{pi}^* and $V_{\theta i}^*$ are bounded, one can have positive values η_p and η_θ with $\|V_{pi}^*\| \leq \eta_p$ and $\|V_{\theta i}^*\| \leq \eta_\theta$. Then, by integrating (38), one can have

$$\begin{aligned} &\int_{t_0}^t e_{pi}^T Q_{pi} e_{pi} d\tau + \int_{t_0}^t e_{\theta i}^T Q_{\theta i} e_{\theta i} d\tau \leq \\ &L(V_{pi}^*(t_0), V_{\theta i}^*(t_0), \tilde{u}_{\Delta pi}(t_0), \tilde{u}_{\Delta \theta i}(t_0)) + \eta_p \int_{t_0}^t \alpha_{1i}(\tau) d\tau \\ &- L(V_{pi}^*(t), V_{\theta i}^*(t), \tilde{u}_{\Delta pi}(t), \tilde{u}_{\Delta \theta i}(t)) \\ &+ \eta_\theta \int_{t_0}^t \alpha_{2i}(\tau) d\tau + \frac{1}{4} (8 + \tilde{u}_{\Delta pi}^2) \int_{t_0}^t \sigma_1(\tau) d\tau \\ &+ \frac{1}{4} (8 + \tilde{u}_{\Delta \theta i}^2) \int_{t_0}^t \sigma_2(\tau) d\tau. \end{aligned} \quad (40)$$

Then, one can obtain from (39), (40) that

$$\begin{aligned} &\int_{t_0}^\infty e_{pi}^T Q_{pi} e_{pi} d\tau + \int_{t_0}^\infty e_{\theta i}^T Q_{\theta i} e_{\theta i} d\tau \leq \\ &L(V_{pi}^*(t_0), V_{\theta i}^*(t_0), \tilde{u}_{\Delta pi}(t_0), \tilde{u}_{\Delta \theta i}(t_0)) + \eta_p \bar{\alpha}_{1i} + \eta_\theta \bar{\alpha}_{2i} \\ &+ \frac{1}{4} (8 + \tilde{u}_{\Delta pi}^2) \bar{\sigma}_1 + \frac{1}{4} (8 + \tilde{u}_{\Delta \theta i}^2) \bar{\sigma}_2 \end{aligned} \quad (41)$$

and

$$\begin{aligned} &L(V_{pi}^*(t), V_{\theta i}^*(t), \tilde{u}_{\Delta pi}(t), \tilde{u}_{\Delta \theta i}(t)) \leq \\ &L(V_{pi}^*(t_0), V_{\theta i}^*(t_0), \tilde{u}_{\Delta pi}(t_0), \tilde{u}_{\Delta \theta i}(t_0)) \\ &+ \eta_p \int_{t_0}^t \alpha_{1i}(\tau) d\tau + \eta_\theta \int_{t_0}^t \alpha_{2i}(\tau) d\tau \\ &+ \frac{1}{4} (8 + \tilde{u}_{\Delta pi}^2) \int_{t_0}^t \sigma_1(\tau) d\tau + \frac{1}{4} (8 + \tilde{u}_{\Delta \theta i}^2) \int_{t_0}^t \sigma_2(\tau) d\tau. \end{aligned} \quad (42)$$

It can be observed from (42) that the tracking errors e_{pi} , $e_{\theta i}$ and the estimation errors $\tilde{u}_{\Delta pi}$, $\tilde{u}_{\Delta \theta i}$ are bounded. From (41) and the Barbalat Lemma [30], one can obtain that $\lim_{t \rightarrow \infty} \|e_{pi}(t)\| = 0$ and $\lim_{t \rightarrow \infty} \|e_{\theta i}(t)\| = 0$.

4. Simulation results

In this section, a multi-quadrotor system including four quadrotors is constructed as (5) with the parameters as $J_{\theta i} =$

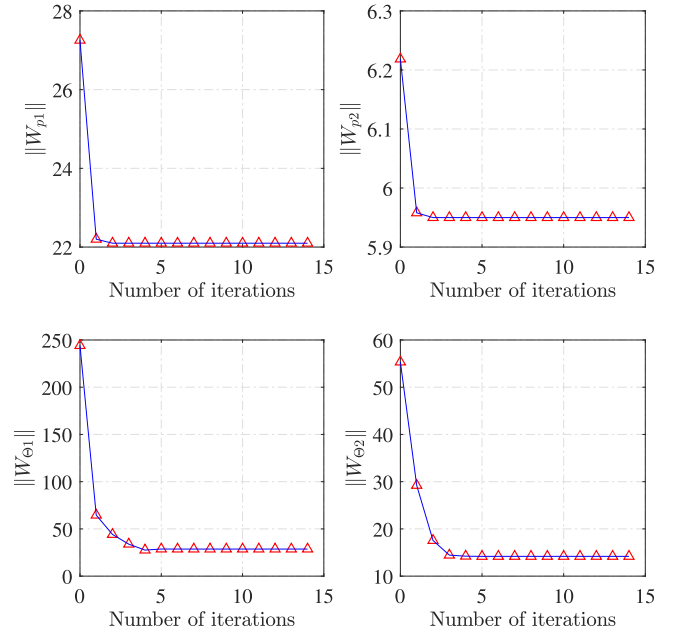


Fig. 1. Convergence of weights in Algorithms 1-2.

$\text{diag}(4, 4, 10) 10^{-3} \text{kg m}^2$, $b_{\theta i} = \text{diag}\{50, 50, 100\}$, $g_i = 9.81 \text{ m/s}^2$, and $b_{pi} = \text{diag}\{1, 1, 1\}$ ($i = 1, 2, 3, 4$). The communication graph G among the quadrotors is set as $w_{14} = w_{21} = w_{32} = w_{43} = 1$ and $\beta_1 = 1$. The relative positions between the quadrotors are set as $\delta_{14} = [1 \ 0 \ 0]^T \text{ m}$, $\delta_{21} = [0 \ -1 \ 0]^T \text{ m}$, $\delta_{32} = [-1 \ 0 \ 0]^T \text{ m}$, and $\delta_{43} = [0 \ 1 \ 0]^T \text{ m}$. The quadrotor team is required to track a straight line with $p_0(t) = [1t \ 0.5t \ 1t]^T \text{ m}$. The initial states of the quadrotor team are set as: $p_1(0) = [1.0 \ 1.0 \ 1.0]^T \text{ m}$, $p_2(0) = [1.0 \ -1.0 \ 0]^T \text{ m}$, $p_3(0) = [-1.0 \ -1.0 \ 0.5]^T \text{ m}$, $p_4(0) = [-1.0 \ 1.0 \ 0.1]^T \text{ m}$, $\dot{p}_i(0) = 0_{3 \times 1}$, $\Theta_i(0) = 0_{3 \times 1}$, and $\dot{\Theta}_i(0) = 0_{3 \times 1}$ ($i = 1, 2, 3, 4$). The scaling factor of the distributed observer satisfies $\chi = 100$. The activation functions of the critic-actor NNs are selected as multiple polynomials with the time interval $\Delta t = 0.05 \text{ s}$. The parameters of the adaptive control laws in (16) and (28) are set as $R_{pi} = R_{\theta i} = I_3$, $\sigma_1(t) = \sigma_2(t) = e^{-t}$, $k_{\sigma_1} = 0.1$, $k_{\sigma_2} = 5$, $\hat{u}_{\Delta pi}(0) = 0.05$, and $\hat{u}_{\Delta \theta i}(0) = 1$. The multiple actuator faults in the rotors of the multi-quadrotor system are set as: $\gamma_{11} = 0.9$, $\gamma_{12} = 0.8$, $\gamma_{13} = 0.7$, $\gamma_{14} = 0.6$, $\gamma_{21} = 0.6$, $\gamma_{22} = 0.7$, $\gamma_{23} = 0.8$, $\gamma_{24} = 0.9$, $\gamma_{31} = 0.9$, $\gamma_{32} = 0.8$, $\gamma_{33} = 0.6$, $\gamma_{34} = 0.9$, $\gamma_{41} = 0.8$, $\gamma_{42} = 0.6$, $\gamma_{43} = 0.7$, and $\gamma_{44} = 0.9$. The occurrence times of the actuator faults of the quadrotors are set as $t_{s1} = 3 \text{ s}$, $t_{s2} = 3.5 \text{ s}$, $t_{s3} = 4 \text{ s}$, and $t_{s4} = 4.5 \text{ s}$. The designing matrices in Algorithms 1-2 satisfy that $Q_{pi} = 10$ and $Q_{\theta i} = 100$. Initial control inputs $u_{ki}^0(t)$ ($k = p, \tau$) with the sum of multiple sinusoidal signals being exploring signals are implemented to generate the quadrotor system data. By using the quadrotor system data, the RL algorithms in Algorithms 1-2 are used to learn the optimal policies u_{pi}^* , $u_{\tau i}^*$ and the convergence of the weights is shown in Fig. 1. It can be seen from Fig. 1 that the convergence of the learning weights is achieved within three iterations. Then, the learned optimal control policies are used to construct the fault-tolerant formation controller for the quadrotor team and the trajectories of the quadrotors in three dimensions (3-D) are shown in Fig. 2. The position tracking errors are shown in Fig. 3 and the attitude tracking errors in Fig. 4.

To provide comparing results, the fault-tolerant methods, including the robust compensating method in [5] and the adaptive

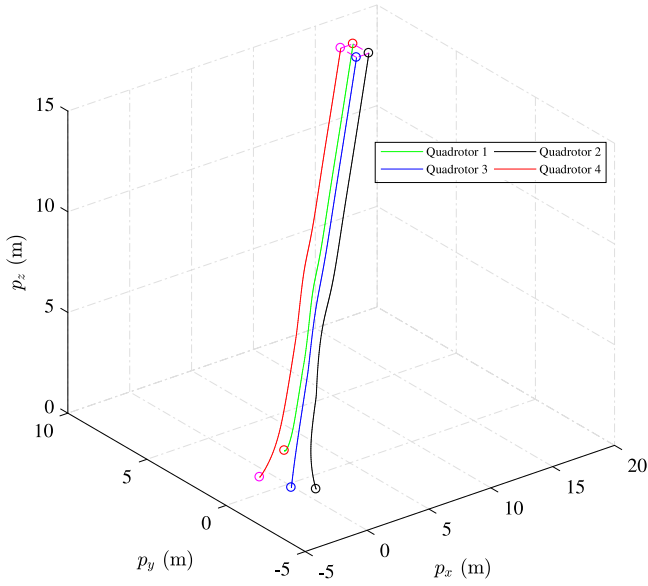


Fig. 2. 3-D trajectories of quadrotor team using the proposed controller.

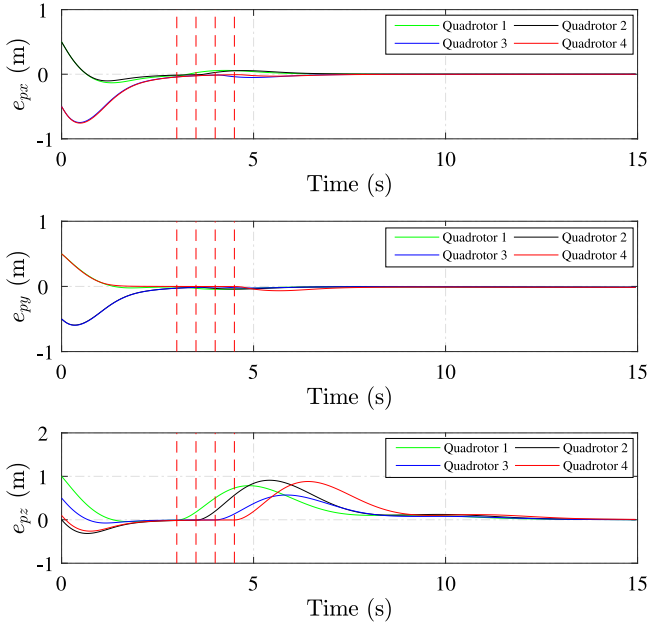


Fig. 3. Position tracking errors of quadrotor team using the proposed controller.

control method in [31], are implemented to the quadrotor systems using the same initial control inputs $u_{ki}^0(t)$ ($k = p, \tau$) in Algorithms 1–2. The position and attitude controller inputs $u_{ki}(t)$ ($k = p, \tau$) using the robust compensating method in [5] are designed as $u_{ki}(t) = u_{ki}^0(t) + u_{ki}^R(t)$, where $u_{ki}^0(t)$ is the initial control input used in Algorithms 1–2 and $u_{ki}^R(t)$ the robust compensating input. $u_{ki}^R(t)$ ($k = p, \tau$) is designed to restrain the effects of the nonlinear dynamics, actuator faults, and parameter uncertainties and satisfy

$$\begin{aligned}\dot{\eta}_{i,p1} &= -f_{ip}\eta_{i,p1} - f_{ip}^2 p_i + b_{pi} u_{pi} - g c_{3,3}, \\ \dot{\eta}_{i,p2} &= -f_{ip}\eta_{i,p2} + 2f_{ip} + \eta_{i,p1}, \\ u_{pi}^R &= b_{pi}^{-1} \eta_{i,p2}^2 (\eta_{i,p2} - p_i),\end{aligned}$$

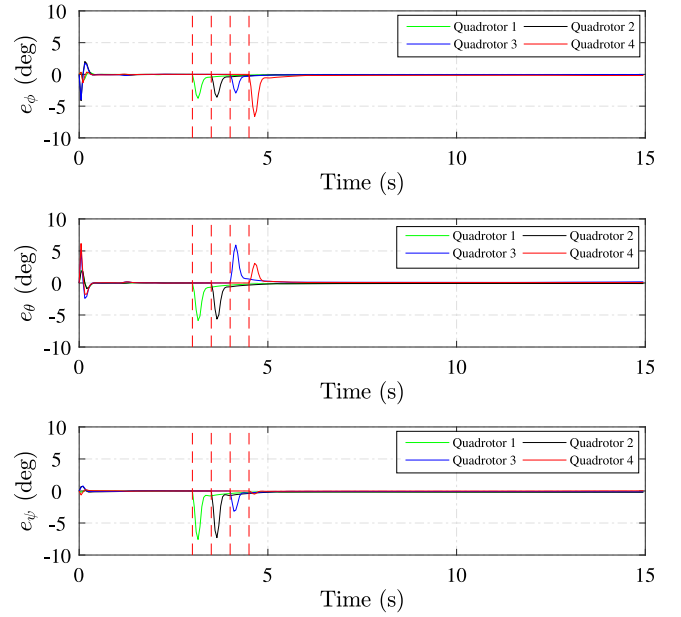


Fig. 4. Attitude tracking errors of quadrotor team using the proposed controller.

and

$$\begin{aligned}\dot{\eta}_{i,a1} &= -f_{ia}\eta_{i,a1} - f_{ia}^2 \Theta_i + b_{\tau i} u_{\tau i} + J_{\Theta i}^{-1} C_i (\Theta_i, \dot{\Theta}_i) \dot{\Theta}_i, \\ \dot{\eta}_{i,a2} &= -f_{ia}\eta_{i,a2} + 2f_{ia} \Theta_i + \eta_{i,a1}, \\ u_{\tau i}^R &= b_{\tau i}^{-1} f_{ia}^2 (\eta_{i,a2} - \Theta_i),\end{aligned}$$

where $f_{ip} \in \mathbb{R}^3$ and $f_{ia} \in \mathbb{R}^3$ are positive robust filter parameters to be determined. Furthermore, based on the initial control input $u_{ki}^0(t)$ ($k = p, \tau$) in Algorithms 1–2, the adaptive control method in [31] are designed as

$$\begin{aligned}u_{pi}(t) &= \frac{1}{2} u_{pi}^0(x) + \frac{\hat{u}_{\Delta pi}^2 R_{pi} u_{pi}^0(x)}{\|R_{pi} u_{pi}^0(x)\| \hat{u}_{\Delta pi} + \sigma_1(t)}, \\ u_{\tau i}(t) &= \frac{1}{2} u_{\tau i}^0(x) + \frac{\hat{u}_{\Delta \Theta i}^2 R_{\Theta i} u_{\tau i}^0(x)}{\|R_{\Theta i} u_{\tau i}^0(x)\| \hat{u}_{\Delta \Theta i} + \sigma_2(t)}.\end{aligned}$$

The parameters for the robust control method in [5] are selected as $\eta_{i,p1} = \mathbf{0}_{3 \times 1}$, $\eta_{i,p2} = \mathbf{0}_{3 \times 1}$, $\eta_{i,a1} = \mathbf{0}_{3 \times 1}$, $\eta_{i,a2} = \mathbf{0}_{3 \times 1}$, $f_{ip} = \text{diag}(1, 1, 1)$, $f_{ia} = \text{diag}(50, 50, 50)$. The parameters for the adaptive control method in [31] are set as the same with the proposed method.

To compare, the initial control inputs $u_{ki}^0(t)$ ($k = p, \tau$) in Algorithms 1–2 are firstly applied to the quadrotor team and the position tracking errors of quadrotor team is shown in Fig. 5. Then, based on the initial control inputs $u_{ki}^0(t)$ ($k = p, \tau$) in Algorithms 1–2, the robust compensating method in [5] is implemented to the quadrotor team with large system parameter uncertainty (the real system parameters are 80% larger than the nominal values) and the position tracking errors of quadrotor team is shown in Fig. 6. The adaptive control method in [31] is applied to the quadrotor team with the same initial control inputs $u_{ki}^0(t)$ ($k = p, \tau$) in Algorithms 1–2 and the position tracking errors of quadrotor team is shown in Fig. 7. One can observe from Figs. 5–7 and Fig. 3 that both the robust compensating control method in [5], the adaptive control method in [31], and the proposed method are able to greatly improve the formation tracking performance of quadrotor team by using the same initial control inputs $u_{ki}^0(t)$ ($k = p, \tau$). However, it can be seen from Fig. 6 that the convergence performance of the robust compensating control method in [5] for quadrotors with large system parameter uncertainty is undesirable. One can observe from Figs. 6–7

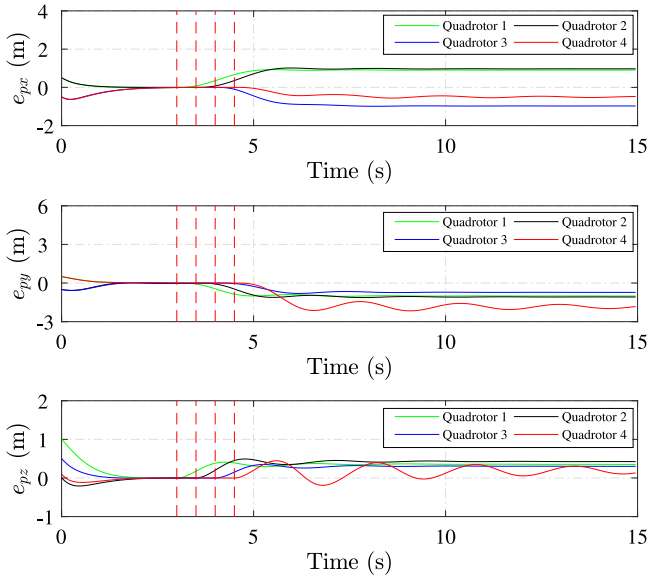


Fig. 5. Position tracking errors of quadrotor team using initial control inputs $u_{ki}^0(t)$ ($k = p, \tau$) in Algorithms 1–2.

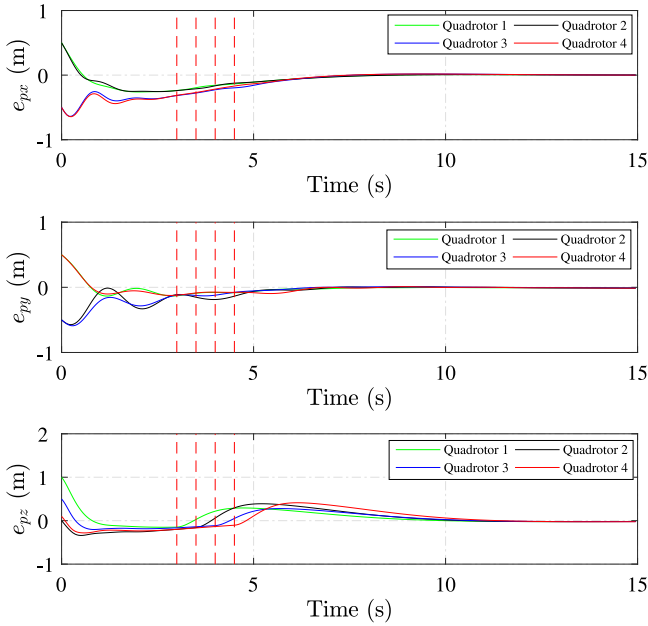


Fig. 6. Position tracking errors of quadrotor team using robust compensating method in [5] with large system parameter uncertainty.

and Fig. 3 that the proposed method shows better convergence performance for quadrotors with unknown system parameters, compared to the robust compensating control method in [5] and the adaptive control method in [31].

5. Conclusion

In this paper, a data-driven fault-tolerant formation control method is provided for quadrotors with nonlinearities, underactuation, and actuator faults in the vehicle dynamics. The multi-quadrotor system is considered as a nonlinear system under

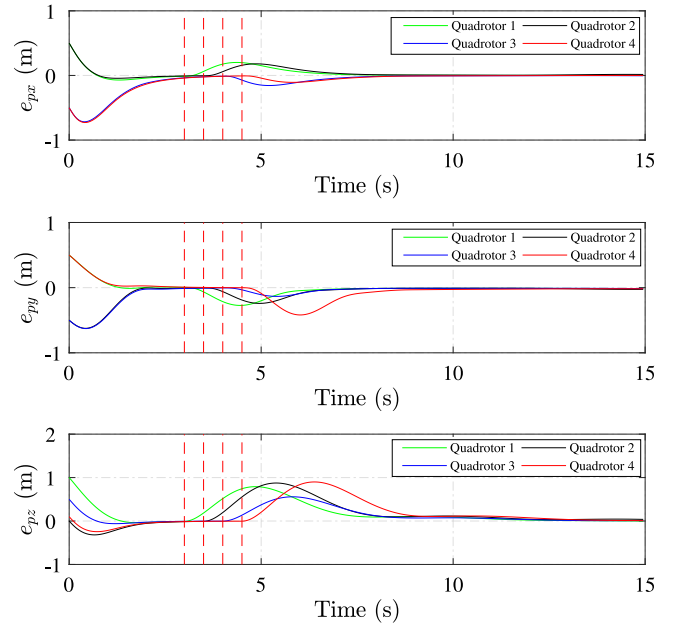


Fig. 7. Position tracking errors of quadrotor team using adaptive control method in [31].

the effects of multiple actuator faults. The fault-tolerant formation controller includes a distributed observer, a position FTC law, and an attitude FTC law. Reinforcement learning approaches are used to update the optimal position and attitude control laws without the knowledge of vehicle dynamics. Stability of the multi-quadrotor system using the proposed control method is discussed and a simulation of quadrotor team under multiple actuator faults is provided to show the effectiveness of the proposed formation controller.

CRedit authorship contribution statement

Wanbing Zhao: Writing - original draft, Formal Analysis, Investigation, Data curation, Software, Validation. **Hao Liu:** Conceptualization, Methodology, Writing - review & editing, Resources, Project Administration, Funding Acquisition. **Yan Wan:** Supervision, Writing - review & editing, Project Administration, Funding Acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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